Mohammadkazem Rajabi

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Summary

AI Engineer with hands-on experience building and shipping **LLM-powered systems** and **agentic AI pipelines** in production. Designed and deployed RAG chatbots, LangGraph ReAct agents, vector-database retrieval systems, and NLP analytics pipelines across real-world veterinary and digital product domains. Strong foundation in machine learning and deep learning from a Master's in AI, complemented by applied thesis work on **LLM-driven code generation** and tool orchestration. Passionate about building reliable, production-grade AI systems end to end.Holder of a Dutch Orientation Year (Zoekjaar) visa — available from June 2026, no sponsorship required

Work Experience

AI Engineer — *Aug 2025 – Present*

Pet 24-7 Global Ltd — *London, United Kingdom*

Live app: App Store - Google Play

Production RAG System

- Built and deployed a **production RAG veterinary chatbot** with role-based logic for pet owners and veterinarians, integrating semantic retrieval, cross-encoder re-ranking, and multilingual query handling over a domain-specific knowledge base.
- Designed a **per-vet episodic memory system** using vector embeddings to store and semantically retrieve past clinical cases, enabling context-aware follow-up responses across sessions.
- Delivered **real-time SSE streaming** and a multimodal document pipeline covering OCR, speech-to-text, and automated clinical PDF report generation.

Agentic AI Platform

- Built the chatbot as a **LangGraph ReAct agent** with pluggable tools for knowledge base retrieval, semantic history search, document OCR, and PDF generation — enabling autonomous multi-step veterinary consultations.
- Engineered a **hybrid memory architecture** combining a PostgreSQL session cache with Milvus vector search for both fast recent-turn recall and long-term semantic history retrieval.

RAG Retrieval Benchmark

- Benchmarked **Milvus, Qdrant, and Weaviate** for production RAG retrieval, using **recursive character chunking** with overlap to preserve semantic coherence, and evaluated across latency, throughput, and retrieval quality on domain-specific queries.

AI Analytics Dashboard

- Built an end-to-end **NLP analytics pipeline** processing all chatbot conversations through a two-layer NER system (domain dictionary + biomedical token classifier) and BERTopic-based topic modeling with LLM-generated labels.
- Developed an **interactive analytics dashboard** exposing entity frequency, trends, co-occurrence, and regional breakdowns across vet and pet-owner flows via a React frontend and FastAPI backend.

Full Stack Developer — *Nov 2023 – Nov 2024*

MeNEW - Personal Digital Menu — *Part-time*

Digital Menu Platform

- Developed and maintained a **full-stack digital menu platform** using **Next.js** for the frontend and **Django** for the backend REST API, delivering a responsive and dynamic user experience.
- Implemented **CI/CD pipelines** to automate testing and deployment, reducing release friction and ensuring consistent delivery across environments.

Technical Skills

Languages: Python, Java, C++, JavaScript

AI/ML: NLP, LLM, RAG, Agentic AI, Deep Learning, Computer Vision

Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FastAPI, Keras, Spark

Libraries: Pandas, NumPy, NLTK, SciPy, BERTopic, Matplotlib, Plotly, Seaborn

Vector Databases: Milvus, Qdrant, Weaviate, ChromaDB

Cloud & Storage: AWS S3, Google Cloud (Vertex AI)

LLM APIs: Google Gemini, OpenAI API, MCP

Web & DevOps: React, Next.js, Node.js, Django, Docker, Git, SQL, MongoDB, PostgreSQL, JWT

Education

University of Padua **Padova, Italy**
*Master in Computer Science (Major in **Artificial Intelligence**)* *Sep 2022 – Sep 2025*
GPA: **105/110**

Thesis: Improving Navigation in LLM-Based AI Frameworks *October 2024 – July 2025*
Built an **agentic framework** where an LLM serves as the reasoning engine, orchestrating domain-specific tools via structured APIs and **in-context prompting**. The LLM generates executable code at runtime to select and compose tools, observes intermediate results, and iterates toward a solution — following a **ReAct-style reasoning loop**. Designed the prompt architecture, extended the tool suite, and benchmarked the system against end-to-end and modular baselines.
Supervisor: Prof. Lamberto Ballan

Course Projects

- **Sentiment and Topic Analysis of Tech Product Reviews** *(Code and report on* GitHub*)*
- **Image Colorization with Conditional GANs and Pretrained U-Net** *(Code and report on* GitHub*)*
- **CNN and Vision Transformer Fine-tuning for Object Detection** *(Code and report on* GitHub*)*
- **LLM Pruning and Fine-Tuning for Efficient Text Summarization**

For other works, explore more on my *GitHub*

Languages

- English: C2 – Studied in English
- Italian: A2
- Persian: Native